EXCELLENCE

CLASS 11 - 2026-27

Physics | Lecture - 17A

Basic Maths and Calculus

Substitution Limits, Area and Average Value

By - Excellence Programme

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Topics to be covered

01 Definite integrals with substitution

02 Changing limits with u

03 Physics examples with substitution

04 Area between two curves

05 Average value of a function

06 Average velocity from v(t)

07 Mixed calculus revision

Definite integrals with substitution

Two clean options: change the limits to u, or substitute back to x then use F(b) − F(a).

Preferred: change limits with u → no need to return to x

If you change variable, you must also change both limits.

Changing limits

When x = a, compute u(a)

When x = b, compute u(b)

Then evaluate $\int_{u(a)}^{u(b)} f(u)\,du$

Do not put x-limits on a u-integral.

Definite with linear u

Evaluate $\int_0^1 (2x + 1)^3 \cdot 2 \, dx$.

Solution: Definite with linear u

Step 1: $u = 2x + 1 \to du = 2\,dx$

Step 2: $x = 0 \to u = 1$; $x = 1 \to u = 3$

Step 3: $\int_1^3 u^3\,du = [u^4/4]_1^3 = 81/4 - 1/4$

20.

Definite sine substitution

Evaluate $\int_0^{(\pi/6)} \cos(3x) \cdot 3\,dx$.

Solution: Definite sine substitution

Step 1: $u = 3x \to du = 3\,dx$

Step 2: $x = 0 \to u = 0$; $x = \pi/6 \to u = \pi/2$

Step 3: $\int_0^{(\pi/2)} \cos u\,du = [\sin u]_0^{(\pi/2)} = 1$

1.

Rough work / extra work

Definite without the factor ready

Evaluate $\int_0^1 e^{2x}\,dx$.

Solution: Definite without the factor ready

Step 1: $u = 2x \to dx = du/2$; $x = 0 \to u = 0$; $x = 1 \to u = 2$

Step 2: $\int_0^2 e^u \, (du/2) = (1/2)[e^u]_0^2 = (1/2)(e^2 - 1)$

$(e^2 - 1)/2$.

Definite reciprocal form

Evaluate $\int_0^1 2x/(x^2 + 1)\,dx$.

Solution: Definite reciprocal form

Step 1: $u = x^2 + 1 \to du = 2x\,dx$

Step 2: $x = 0 \to u = 1$; $x = 1 \to u = 2$

Step 3: $\int_1^2 (du/u) = [\ln|u|]_1^2 = \ln 2 - \ln 1$

$\ln 2$.

Rough work / extra work

Physics with substitution

When acceleration or velocity is a composite function of time, substitute the inside.

Same four steps — then put the physical limits of t

Keep units on the final number.

Velocity from a = 2t + 1

If a = 2t + 1 and v = 0 at t = 0, find v at t = 2.

Solution: Velocity from a = 2t + 1

Step 1: $v = \int_0^2 (2t + 1)\,dt = [t^2 + t]_0^2$

Step 2: $= (4 + 2) - 0 = 6$

$v = 6\ \text{m s}^{-1}$.

Displacement with $v = 3t^2$

If $v = 3t^2$ and $x = 0$ at $t = 0$, find x at $t = 2$.

Solution: Displacement with $v = 3t^2$

Step 1: $x = \int_0^2 3t^2\,dt = [t^3]_0^2 = 8$

$x = 8$ m.

Rough work / extra work

Composite velocity

If $v = 2t\cos(t^2)$ and $x = 0$ at $t = 0$, find x at $t = \sqrt{(\pi/2)}$.

Solution: Composite velocity

Step 1: $u = t^2 \rightarrow du = 2t\,dt$; $t = 0 \rightarrow u = 0$; $t = \sqrt{(\pi/2)} \rightarrow u = \pi/2$

Step 2: $x = \int_0^{(\pi/2)} \cos u\,du = [\sin u]_0^{(\pi/2)} = 1$

$x = 1$ m.

Mixed: identify u

Find $\int 6x^2 (x^3 + 4)^5 \, dx$.

Solution: Mixed: identify u

Step 1: $u = x^3 + 4 \rightarrow du = 3x^2\,dx \rightarrow 6x^2\,dx = 2\,du$

Step 2: $\int 2u^5\,du = 2 \cdot u^6/6 + C = u^6/3 + C$

$(x^3 + 4)^6/3 + C$.

Mixed: definite power

Evaluate $\int_0^1 4x\,(x^2 + 1)^3\,dx$.

Solution: Mixed: definite power

Step 1: $u = x^2 + 1 \to du = 2x\,dx \to 4x\,dx = 2\,du$

Step 2: $x = 0 \to u = 1$; $x = 1 \to u = 2$

Step 3: $\int_1^2 2u^3\,du = 2[u^4/4]_1^2 = (1/2)(16 - 1)$

15/2.

Rough work / extra work

Homework: sin(2x)

Find $\int \sin(2x)\, dx$.

Solution: Homework: sin(2x)

Step 1: $u = 2x \rightarrow dx = du/2$

$-(1/2)\cos(2x) + C.$

Homework: definite e^{3x}

Evaluate $\int_0^1 e^{3x}\,dx$.

Solution: Homework: definite e^{3x}

Step 1: $u = 3x \to dx = du/3$; limits $0 \to 3$

Step 2: $(1/3)[e^u]_0^3 = (e^3 - 1)/3$

$(e^3 - 1)/3$.

Rough work / extra work

Homework: check by differentiate

Differentiate $y = (1/5)\sin(5x)$ and confirm it matches $\cos(5x)$.

Solution: Homework: check by differentiate

Step 1: $dy/dx = (1/5) \cdot \cos(5x) \cdot 5 = \cos(5x)$

Check successful — so $\int \cos(5x)\,dx = (1/5)\sin(5x) + C$.

Substitution lock lines

Substitution = chain rule written backwards

Choose u, write du, rewrite the whole integral

For definite integrals, change both limits with u

Always check by differentiating the answer

Practice

1. Redo every substitution question without looking.

2. For each indefinite answer, differentiate and recover the integrand.

3. Practise changing limits on three definite examples.

4. Write one physics $\int$ with a composite function of t.

Leaving a mix of u and x in the same integral is the classic trap.

Substitution recap

Choose u, write du, rewrite the whole integral

Definite: change both limits with u

$\int_a^b f = F(b) - F(a)$ (no +C)

Check by differentiating

Quick check: substitution

Find $\int 2x \cos(x^2)\, dx$.

Solution: Quick check: substitution

Step 1: $u = x^2 \rightarrow du = 2x\,dx$

$\sin(x^2) + C$.

Rough work / extra work

Area between two curves

When one curve lies above another from x = a to x = b, the area between them is the difference of integrals.

$$\text{Area} = \int_a^b [y_{\text{upper}} - y_{\text{lower}}]\, dx$$

Always upper minus lower. Do not subtract in the wrong order.

Three board steps

1. **Sketch both curves and mark a and b**

2. **Decide which is upper on [a, b]**

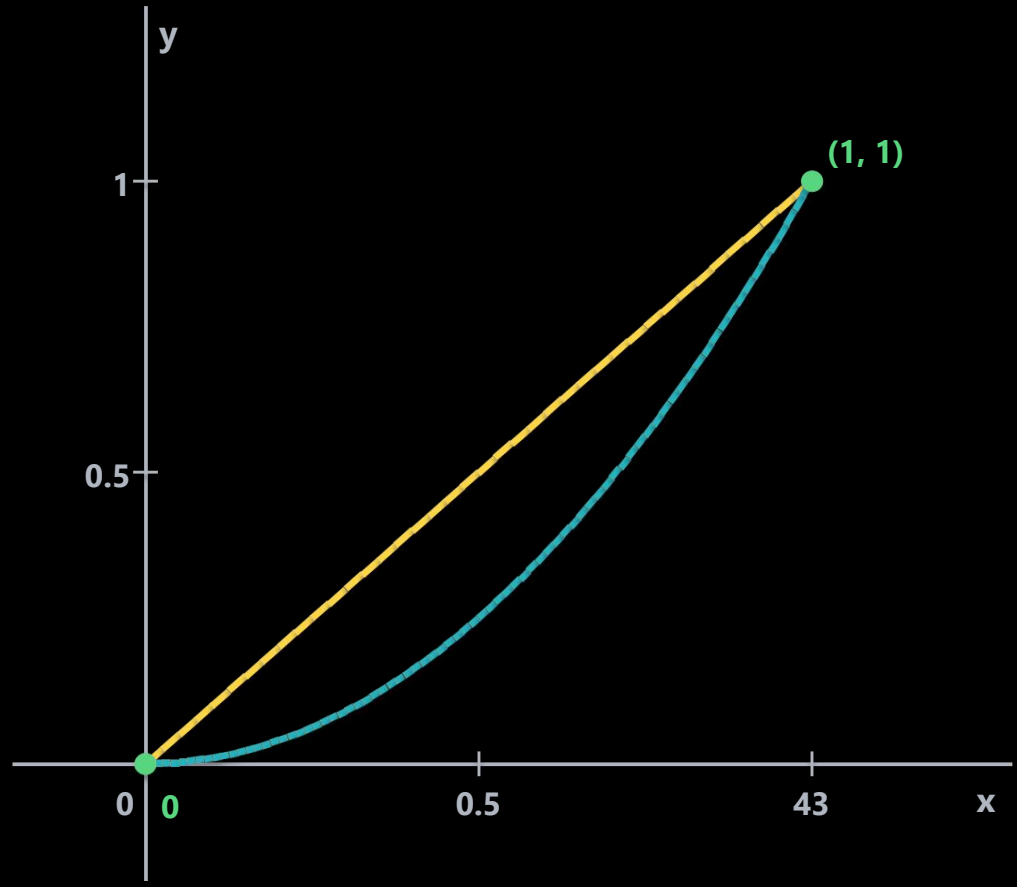
3. **Integrate (upper − lower) from a to b**

If they cross, split the integral at the intersection.

Region between $y = x$ and $y = x^2$

On [0, 1], the line $y = x$ is above the parabola $y = x^2$.

$$\text{Area} = \int_0^1 (x - x^2)\,dx$$



Area between line and parabola

Find the area between $y = x$ and $y = x^2$ from $x = 0$ to $x = 1$.

Solution: Area between line and parabola

Step 1: Upper = x, lower = x^2

Step 2: Area = $\int_0^1 (x - x^2)\,dx = [x^2/2 - x^3/3]_0^1$

Step 3: = 1/2 − 1/3 = 1/6

Area = 1/6 square units.

Rough work / extra work

Area between line and axis

Find the area between $y = 2x$ and the x-axis from $x = 1$ to $x = 3$.

Solution: Area between line and axis

Step 1: This is still $\int_1^3 2x\,dx$ (upper = 2x, lower = 0)

Step 2: $= [x^2]_1^3 = 9 - 1 = 8$

Area = 8 square units.

Area between two lines

Find the area between $y = 4$ and $y = x$ from $x = 1$ to $x = 3$.

Solution: Area between two lines

Step 1: On [1, 3], y = 4 is above y = x

Step 2: Area = $\int_1^3 (4 - x)\,dx = [4x - x^2/2]_1^3$

Step 3: = (12 − 9/2) − (4 − 1/2) = 15/2 − 7/2 = 4

Area = 4 square units.

Find intersection first

Find the area between $y = x^2$ and $y = 4x - x^2$.

Solution: Find intersection first

Step 1: Intersection: $x^2 = 4x - x^2 \rightarrow 2x^2 - 4x = 0 \rightarrow 2x(x - 2) = 0$

Step 2: $x = 0$ or $x = 2$. On $[0, 2]$, upper $= 4x - x^2$

Step 3: Area $= \int_0^2 [(4x - x^2) - x^2]\, dx = \int_0^2 (4x - 2x^2)\, dx$

Step 4: $= [2x^2 - (2/3)x^3]_0^2 = 8 - 16/3 = 8/3$

Area = 8/3 square units.

Rough work / extra work

Average value of a function

The average height of y = f(x) on [a, b] is the area under the curve divided by the width.

$$f_{avg} = (1/(b - a)) \int_a^b f(x)\, dx$$

This is the mean value of f on that interval.

Do not forget to divide by (b − a).

Why this formula?

Area under f equals the area of a rectangle of width (b − a) and height f_avg.

$$\int_a^b f(x)\,dx = f_avg \cdot (b - a)$$

Solve for f_avg to get the formula.

Average of a line

Find the average value of f(x) = 2x on [0, 4].

Solution: Average of a line

Step 1: $f_{avg} = (1/4) \int_0^4 2x\,dx = (1/4)[x^2]_0^4$

Step 2: $= (1/4)(16) = 4$

Average value = 4.

Rough work / extra work

Average of a parabola

Find the average value of $f(x) = x^2$ on $[0, 3]$.

Solution: Average of a parabola

Step 1: $f_{avg} = (1/3)\int_0^3 x^2\,dx = (1/3)[x^3/3]_0^3$

Step 2: $= (1/3)(9) = 3$

Average value = 3.

Average of cosine

Find the average value of $f(x) = \cos x$ on $[0, \pi/2]$.

Solution: Average of cosine

Step 1: Width = $\pi/2 - 0 = \pi/2$

Step 2: $f_{avg} = (2/\pi) \int_0^{(\pi/2)} \cos x \, dx = (2/\pi)[\sin x]_0^{(\pi/2)}$

Step 3: $= (2/\pi)(1) = 2/\pi$

Average value = $2/\pi$.

Physics: average velocity

If v(t) is given, the average velocity over $[t_1, t_2]$ is the mean value of v.

$$v_{avg} = (1/(t_2 - t_1)) \int_{t1}^{t2} v(t)\, dt = \Delta x / \Delta t$$

Same formula as average value — now with time as the variable.

Rough work / extra work

Average velocity from v(t)

If $v = 2t$ (m s^{-1}) from $t = 0$ to $t = 4$ s, find average velocity.

Solution: Average velocity from v(t)

Step 1: $v_{avg} = (1/4)\int_0^4 2t\,dt = (1/4)[t^2]_0^4 = 4$

Step 2: Check: $\Delta x = 16$ m, $\Delta t = 4$ s $\rightarrow v_{avg} = 4$ m s^{-1}

$v_{avg} = 4$ m s^{-1}.

Average of constant acceleration speed-up

If $v = 3 + 2t$ from $t = 0$ to $t = 3$, find v_{avg}.

Solution: Average of constant acceleration speed-up

Step 1: $v_{avg} = (1/3)\int_0^3 (3 + 2t)\,dt = (1/3)[3t + t^2]_0^3$

Step 2: $= (1/3)(9 + 9) = 6$

$v_{avg} = 6\ m\ s^{-1}$.

Mean of a linear a

If a = 4t from t = 1 to t = 3, find average acceleration.

Solution: Mean of a linear a

Step 1: $a_{avg} = (1/2)\int_1^3 4t\,dt = (1/2)[2t^2]_1^3$

Step 2: $= (1/2)(18 - 2) = 8$

$a_{avg} = 8\ m\ s^{-2}$.

Rough work / extra work

Indefinite integral lock lines

$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$

$\int \sin x\,dx = -\cos x + C \cdot \int \cos x\,dx = \sin x + C$

$\int e^x\,dx = e^x + C \cdot \int \frac{dx}{x} = \ln|x| + C$

Substitution undoes the chain rule

Revision: indefinite

Find $\int (4x^3 - 1/x)\,dx$ for $x > 0$.

Solution: Revision: indefinite

Step 1: $\int 4x^3\,dx = x^4$

Step 2: $\int -1/x\,dx = -\ln x$

$x^4 - \ln x + C$.

Revision: definite

Evaluate $\int_0^{(\pi/4)} \cos(2x) \cdot 2\,dx$.

Solution: Revision: definite

Step 1: $u = 2x \to du = 2\,dx$; limits $0 \to \pi/2$

Step 2: $\int_0^{(\pi/2)} \cos u\,du = 1$

1.

Rough work / extra work

Revision: area between

Find the area between $y = \sqrt{x}$ and $y = x$ from $x = 0$ to $x = 1$.

Solution: Revision: area between

Step 1: On [0, 1], $\sqrt{x} \geq x$

Step 2: Area = $\int_0^1 (\sqrt{x} - x)\,dx = \left[(2/3)x^{3/2} - x^2/2 \right]_0^1$

Step 3: = 2/3 − 1/2 = 1/6

Area = 1/6 square units.

Revision: average value

Find the average value of $f(x) = 3x^2$ on $[1, 2]$.

Solution: Revision: average value

Step 1: $f_{avg} = (1/1) \int_1^2 3x^2\,dx = [x^3]_1^2 = 8 - 1$

Average value = 7.

Rough work / extra work

Homework: between curves

Find the area between $y = 6 - x$ and $y = x$ from $x = 1$ to $x = 2$.

Solution: Homework: between curves

Step 1: Upper = 6 − x on this interval

Step 2: Area = $\int_1^2 (6 - x - x)\,dx = \int_1^2 (6 - 2x)\,dx$

Step 3: $= [6x - x^2]_1^2 = (12 - 4) - (6 - 1) = 3$

Area = 3 square units.

Homework: average velocity

If $v = t^2$ from $t = 0$ to $t = 3$, find v_{avg}.

Solution: Homework: average velocity

Step 1: $v_{avg} = (1/3)\int_0^3 t^2\,dt = (1/3)[t^3/3]_0^3 = 3$

$v_{avg} = 3\ m\ s^{-1}$.

Calculus block lock lines

Area between curves: $\int$ (upper − lower) dx

Average value: $f_{avg} = (1/(b-a)) \int_a^b f\, dx$

$v_{avg} = (1/\Delta t) \int v\, dt = \Delta x/\Delta t$

Differentiation ↔ Integration are reverse operations

Rough work / extra work

Practice

1. Sketch every area-between-curves question before integrating.

2. Memorise the average-value formula with (b − a) in the denominator.

3. Redo one substitution + one definite + one average question.

4. Write a one-page summary of the whole calculus block.

Next chapter after this block: Vectors.